Unsupervised Learning Project: A Reproducible Experimental Study of Clustering

Overview

This semester-long project requires each team to conduct a complete, reproducible clustering study on a real-world dataset provided for the course. The work must be framed as a clearly specified data analysis question and executed as an end-to-end data mining process: data understanding and preprocessing → (optional) representation learning → clustering → evaluation and robustness analysis → interpretation.

The focus is on a principled, reproducible study with clearly justified choices. Teams must (i) justify modelling choices, (ii) define a principled and reproducible evaluation protocol, (iii) provide evidence of sensitivity and stability, and (iv) draw conclusions that are supported by empirical results. The report should state explicitly what is being measured, how it is modelled, how quality is assessed, and whether the findings answer—or fail to answer—the formulated question.

Project Objectives

- Formulate and motivate a clustering problem to be addressed without class labels.
- Design and implement a reproducible data mining pipeline for mixed-type tabular data.
- Select, justify, and compare clustering models (**mandatory**: k-means baseline + at least one additional family).
- Define and justify "what to measure" (validity indices, stability, and problem-relevant proxies), and use them consistently.
- Interpret results in context by profiling clusters, stating limitations, and deriving evidence-based, decision-relevant recommendations (or testable hypotheses) supported by evidence.

Research Questions (to be answered in the report)

- RQ1 — ***Problem framing***: What is the concrete segmentation question, and why is clustering an appropriate tool?
- RQ2 — ***Representation***: Which preprocessing/encoding choices for mixed data are justified, and how do they affect similarity and clustering?
- RQ3 — ***Modelling***: Which clustering family and configuration yields the most coherent, and stable clusters under the chosen protocol?
- RQ4 — ***Interpretation***: What do the discovered clusters represent, and do they provide useful information to address the original question?
- RQ5 **— Robustness:** How sensitive are the conclusions to reasonable variations in preprocessing, representation, and model configuration (e.g., scaling choices, hyperparameter search space, and randomness/resampling)?

Teams

- **Team size:** 3 students.
- **Submission:** deliverables (report + repository) are submitted per team.
- **Individual assessment:** a base grade is assigned to the team deliverables (technical quality, methodology, evidence, reproducibility). Individual grades may be adjusted based on the oral discussion and the self/peer assessment, considering demonstrated understanding and effective contribution.

Dataset

Mandatory dataset: *Hotel Booking Demand* (tabular, mixed numerical/categorical attributes**):**

- **Available at**: https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand/data
- **Reference**: António, N., de Almeida, A., & Nunes, L. (2019). *Hotel booking demand datasets*. Data in Brief, 22, 41–49. https://doi.org/10.1016/j.dib.2018.11.126
- **Unit of analysis:** each row corresponds to one booking record.
- **Course release:** the dataset will be provided through the course platform in a fixed version. Teams must reference the dataset version used (e.g., "course release v1") and include a checksum/hash in the dataset documentation to ensure traceability.
- **Governance:** the report must include the dataset's usage terms/license (as stated in the release) and any relevant restrictions. The raw dataset must **not** be committed to the repository; instead, include instructions/scripts to obtain and prepare it.

Experimental requirements

- Leakage control: outcome/post-event variables must not be used to form clusters. At minimum, exclude: is_canceled and reservation_status (and any other post-arrival or post-decision attributes, depending on the chosen scenario). Outcome/post-event variables may be used only for post-hoc profiling/interpretation, not as clustering inputs.
- Remove ID-like or extremely high-cardinality fields (e.g., agent/company identifiers), unless there is a clear justification and a responsible encoding strategy.
- Hyper-parameter intervals must be predefined and justified. Unbounded searches are not permitted.
- If subsampling is required for computing, define a fixed rule (sample size + random seed) and report its impact.
- High-cardinality categorical variables: unless justified, handle high-cardinality fields by dropping them, grouping rare categories into "Other", and/or applying a minimum-frequency threshold prior to one-hot encoding. Any alternative strategy must be justified and its impact discussed.
- Distance/metric governance: teams must state the distance/similarity used by each model (and any changes induced by preprocessing/representation) and ensure that evaluation indices are computed consistently in the same representation/metric space.

Tasks

Task 1— Baseline Study

1.1 Data preparation and validation pipeline

- **State the concrete segmentation question (RQ1) and the unit of analysis**, then produce a short data documentation: source, license/terms, what each row represents, time span (if applicable), and known data quality issues.
- Report missingness and outliers by attribute type (numerical vs categorical).
- Build an end-to-end preprocessing pipeline (may explore: scikit-learn ColumnTransformer + Pipeline):
  - Separate numerical vs categorical features explicitly.
  - Handle missing data with type-appropriate strategies (and rationale).
  - Encode categorical variables (baseline: one-hot; handle unknown categories).
  - Scale numerical variables (standardize).
- Define the final feature set **and the similarity/distance implied by the final representation**; document all exclusions (especially leakage control).

1.2 Learn a baseline clustering model

- Run *k*-means (or MiniBatch k-means) as the baseline method (mandatory).
- Select a small, justified range of k values; do not use unbounded searches.
- Provide an initial interpretation of the resulting clusters (cluster profiles).
- The chosen range of *k* must be justified (e.g., domain plausibility, internal-index behaviour, and interpretability), and the final selection rule must be explicitly stated.
- Include *iK-means* and evaluate it under the same protocol (runtime + validation metrics).

Task 2— Alternative Clustering Family

2.1 Development

- Choose one additional, conceptually distinct clustering family and implement it on the same representation:
  - Agglomerative hierarchical clustering (Ward/average/complete; distance must match the representation), or
  - Gaussian Mixture Model (EM) with model selection via AIC/BIC, or
  - Density-based clustering (DBSCAN) with correct parameter interpretation.
- Document the modelling choices and the final selection rule (parameters and k, when applicable).
- **Note**[1]
  - **Hierarchical (Ward):** Ward linkage requires Euclidean geometry; use only if representation/ distance is compatible.
  - **DBSCAN/OPTICS:** report the fraction of points labelled as noise and explain how noise is handled in evaluation and interpretation.

[1] Discuss with the Professor.

2.2 Evaluation

- Compare the baseline (Task 1.2) against the second family using the evaluation protocol defined in Task 3.

Task 3— Evaluation and Robustness Analysis

The following evaluation protocol is used consistently to compare the baseline (Task 1) and the alternative family (Task 2).

3.1 Internal validity indices

- Report Silhouette score and at least one additional internal index (Calinski–Harabasz or Davies–Bouldin).
- All indices must be computed in the same representation and metric space used for clustering to the dataset.
- If a model uses a non-Euclidean distance or a precomputed distance matrix, indices must be computed in a consistent manner (or a justified alternative must be provided). The chosen distance/space must be clearly stated.

3.2 Family-specific diagnostics

- Include at least one diagnostic appropriate to the selected model family:
  - EM: AIC/BIC curves.
  - DBSCAN/OPTICS: k-distance or reachability plots and a parameter justification.
  - Hierarchical: dendrogram and a rationale for the cut (or linkage choice).

3.3 Stability and sensitivity analysis

- Repeat the full pipeline for at least 5 random seeds (or bootstrap/resampling splits where applicable).
- Report mean ± standard deviation of internal indices.
- Include at least one stability measure (e.g., ARI between runs) and interpret the results.
- Include at least one controlled preprocessing variant (e.g., StandardScaler vs RobustScaler[2]; with vs without PCA) and report the impact.
- For deterministic methods (e.g., hierarchical clustering with fixed linkage and distance), replace "random seeds" with controlled resampling/perturbation (e.g., bootstrap resamples or subsample-and-recluster) and report variability accordingly.

3.4 Interpretation and communication

- Provide cluster profiles with descriptive statistics and visual summaries.
- Indices alone are insufficient: link the clustering result to a meaningful "booking profile" narrative **that addresses the segmentation question**, supported by evidence.

Task 4— Reporting and Reproducibility

4.1 Report

- Write an academic-style report: 6–8 pages (single-column), excluding references and appendix.
- Recommended sections: Abstract; Introduction; Dataset documentation; Methods; Experiments; Results; Discussion; Limitations/Ethics; References.
- The Discussion/Conclusion must explicitly address: (i) what was measured and why; (ii) how modelling choices were made to measure it (representation, similarity, models); (iii) which evaluation protocol was

[2] Standard normalizations / median–IQR scaling (outlier-robust).

followed; and (iv) how results should be interpreted in the context of the formulated problem (including limitations).

4.2 Repository and trace

- Maintain a machine-readable experiment log (e.g., experiments.csv) with method, parameters, seeds, metrics, diagnostics, and notes.
- Include an environment specification (environment.yml or lockfile) and a single-entry point (run_all) that regenerates all results.
- Every figure and table in the report must be reproducible from the repository without manual steps.
- **Repository structure (minimum):** README.md, environment.yml (or lockfile), run_all entry point, experiments.csv, and folders src/, figures/, tables/.
- **Reproducibility check:** run_all must regenerate every figure/table reported, end-to-end, without manual steps. If full runs are computationally heavy, provide a documented "fast check" mode that verifies the pipeline integrity.

Extension Modules

The project grade combines two parts: Core (0–13 points) and Extensions (0–7 points), for a maximum of 20 points. Teams can obtain up to 13/20 using the Core tasks only. Grades above 13/20 require one or more Extension Modules and these must be integrated into the main experimental protocol and report narrative (not placed as an isolated appendix).

Rules

- Extensions must be methodologically correct, fully reproducible under the same standards as the Core, and explicitly linked to the research questions.
- Extension experiments must follow the same project requirements (e.g., leakage control, feature governance, hyper-parameter ranges, and consistent distance/metric reporting).
- If an extension involves randomness (e.g., k-NN graph construction, initializations), include a stability check with at least 3 seeds and report variability (e.g., mean ± standard deviation of key metrics).

Module E1 — Cluster-aware anomaly analysis

Goal: identify and interpret unusual bookings as observations that are rare with respect to the learned cluster structure (i.e., anomalies are defined relative to clusters rather than as a separate anomaly-detection project).

- Define an anomaly score conditioned on clusters (choose at least one): distance to centroid/medoid (normalised by cluster dispersion), low likelihood under a GMM, or points labelled as noise by DBSCAN/OPTICS.
- Report a small and auditable set (e.g., top-20 anomalies), including a brief justification for each (why it is unusual).
- Distinguish data-quality anomalies (inconsistencies, impossible values) from rare-but-plausible behaviour.
- Sensitivity: show how the anomaly set changes (or remains stable) under at least one controlled representation/model change (e.g., scaling choice; with vs. without PCA; k-means vs. GMM).
- Discuss practical relevance and limitations (no ground-truth anomaly labels; risk of conflating "rare" with "bad").

Module E2 — Spectral / graph clustering

Goal: apply spectral clustering with a complete and well-specified graph construction and compare it against at least one Core method under the same evaluation protocol.

- Graph construction: specify k-NN vs mutual-kNN, the value(s) of k, and the distance used in the chosen representation.
- Edge weights: use either connectivity weights (0/1) or a heat-kernel weighting scheme; justify any bandwidth parameter (σ) if used.
- Laplacian choice: specify the Laplacian (unnormalised or normalised) and the eigen-solver configuration.
- Embedding and clustering: state the number of eigenvectors used (eigengap or fixed rule), any embedding normalisation (e.g., row-normalisation), and the final clustering method in the embedding (typically k-means).
- Robustness: include a small sensitivity analysis on k (graph) and/or the weighting choice, and comment on stability and interpretability.

Module E3 — Fuzzy clustering and fuzzy validity

Goal: model soft cluster memberships (fuzzy c-means) and discuss membership uncertainty using fuzzy-appropriate validity criteria.

- Run fuzzy c-means and report membership distributions; interpret at least one case of ambiguous membership (soft assignments).
- Include at least one fuzzy validity measure (e.g., xie-beni, partition coefficient, classification entropy), and explain what it measures.
- Compare the fuzzy solution to a hard clustering baseline (e.g., k-means) in terms of clustering interpretation and stability.

Module E4 — PCA/SVD study for clustering

Goal: study representation learning for clustering (linear DR) and provide evidence for (or against) its usefulness on this dataset.

- Choose a dimensionality-reduction approach (PCA on dense features or TruncatedSVD on sparse one-hot features) and justify the retained dimensionality (e.g., explained variance threshold).
- Compare clustering results in the original representation vs the reduced representation, using the same evaluation and stability protocol.
- Include the key methodological caveat: variance preservation is not guaranteed to preserve cluster-separating directions; support your conclusion with empirical evidence.

Module E5 — Nonlinear DR for visualization & embedding reliability

Goal: use nonlinear dimensionality reduction to support interpretation (visualisation), while explicitly discussing embedding reliability.

- Use one method (MDS/Isomap/LLE/t-SNE/UMAP) for visualisation only. Do not present clustering performed solely on the nonlinear embedding as the main result.
- Report a trustworthiness-style diagnostic (or a parameter sensitivity study) and discuss how conclusions depend on hyperparameters.
- Use the visualisation to support the interpretation of clusters obtained in the primary clustering space.

Project Deliverables and Timeline

The project includes **three mandatory deliverables**, aligned with the practical sessions:

Date / Lab	Checkpoint and Required Artefacts	Focus
Fri 20 Mar / after Lab P4	**Pitch:** problem statement; dataset and documentation (draft); success criteria; risks/ethics considerations; repository scaffold.	Feasibility; data governance; initial design.
Fri 24 Apr / after Lab P8	**Mid-term Technical Progress:** completed EDA; at least two clustering baselines with internal indices; defined sensitivity plan; experiment logging in place.	Technical correctness; early robustness evidence.
Friday 5 Jun 2026, 23:59	**Final Submission:** academic report (6–8 pages, single column) and reproducible repository.	Complete methodology; evaluation; interpretation; reproducibility.

Intermediate Milestones

The following internal milestones support methodological discipline and reproducibility. No separate deliverable is required unless otherwise specified.

Date / Lab	Milestone	Focus
Wed 13 May / Lab P11	**Core Protocol Freeze —** modelling choices and parameter ranges fixed; seeds locked; draft result tables/figures prepared.	Principled selection; comparability.
Wed 20 May / Lab P12	**Reproducibility Freeze —** environment/lockfile finalised; run_all executes end-to-end; tables/figures generated automatically.	Reproducibility; documentation.
Wed 27 May / Lab P13	**Final Readiness Review —** short rehearsal; Q&A preparation; checklist close-out.	Communication; readiness.

GitHub repository

Tag the submission code version as Submission (git tag or GitHub release) before the deadline. Do not commit the raw dataset to the repository. Provide instructions to obtain it (and any preprocessing scripts).
Maintain a clean commit history and keep experiments traceable (e.g., experiments.csv + recorded seeds).

Submission

Teams must ensure consistency between the submitted report and the submitted repository version.

Peer Review

Each student will submit a short self/peer assessment (form will be provided).

Peer review is used to support individual grading and to resolve discrepancies in contribution within a team.

Academic Integrity and Use of Tools

- All code, analyses, figures, and report text must be produced by the team. Any external material (code, text, figures, or datasets) must be properly cited and used in accordance with its license.
- Generative-AI tools (e.g., ChatGPT, Copilot) may be used **solely for consultation purposes** (e.g., clarifying concepts, reviewing syntax, or discussing methodological alternatives). They must **not** be used to generate substantive code, figures, analyses, or report text. Any consultation with such tools must be explicitly disclosed in the report or repository (briefly stating the purpose of use).
- Any use of generative-AI tools beyond consultation, such as producing code, written text, analyses, or experimental results, is considered academic misconduct and will result in failure of this project component.
- During the oral discussion, teams must be able to clearly explain and reproduce any part of their work. Inability to justify modelling decisions, reproduce results, or explain implementation details may result in the oral assessment grade replacing the project grade, in accordance with course regulations.

Recommendation Notes

4. Leakage control

Leakage control means preventing outcome information (or post-event information) from entering the feature set used to form clusters. Clustering inputs must be restricted to variables that are available at the time you intend to assign a booking to a segment.

4.1 Practical implications

- At minimum, exclude outcome/post-event fields such as is_canceled and reservation_status from the clustering inputs.
- Outcome/post-event variables may be retained only for post-hoc profiling and interpretation (e.g., comparing cancellation rates across clusters).
- Document exclusions explicitly and justify them (why they leak outcome or are post-decision/post-arrival).

4.2 Documentation template (example)

- Excluded is_canceled: outcome variable; not available at *index time*.
  Index time (segmentation time) = the time at which a booking would be assigned to a cluster; clustering features must be available **at or before** this time.
- Excluded reservation_status: post-event booking status; realised after booking.
- If other attributes are post-arrival/post-decision in your chosen scenario, list them similarly and justify the exclusion.

5. Feasibility: Ward hierarchical and DBSCAN/OPTICS

The Hotel Booking Demand dataset is mixed-type (numerical + categorical). Feasibility therefore depends on your representation and the metric implied by that representation, as well as computational scale.

5.1 Hierarchical clustering (Ward linkage)

- Ward linkage assumes Euclidean geometry (it minimises within-cluster variance under Euclidean distance). Use Ward only if your final representation is Euclidean-compatible.
- Compatible example: one-hot encode categorical variables + scale numeric variables, then apply Euclidean distance in the resulting vector space.

- Not compatible: Ward is not appropriate if you rely on a non-Euclidean mixed-type distance (e.g., Gower) or other non-Euclidean similarity.
- Practical caveat: agglomerative hierarchical methods often scale as O(n^2) in time/memory; on large n you may need a fixed, reported subsampling rule (size + seed) and you must report the impact.

5.2 DBSCAN / OPTICS

- DBSCAN/OPTICS are permissible alternative families, but density depends strongly on the representation.
- In high-dimensional sparse one-hot spaces, distances can become less informative; DBSCAN may yield one large cluster with many noise points, or almost all noise, unless parameters and representation are chosen carefully.
- Required reporting: always report the fraction of points labelled as noise.
- Define how noise is handled in evaluation (e.g., compute internal indices on non-noise points and report coverage = 1 – noise fraction) and in interpretation (treat noise as 'unassigned/atypical' and discuss whether it reflects data-quality issues or rare-but-plausible behaviour).
- Computational note: for large datasets, neighbor-search may be expensive; if subsampling is required, use a fixed rule and report its impact.